

Daily Extreme-Weather Features for Interpretable Crop-Yield Prediction and Crop-Region Suitability Analysis

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Abstract

Crop-yield models are often evaluated mainly by predictive accuracy, but agricultural planning also requires interpretable information about which crops perform better, where they perform better, and under what weather conditions. This study develops a daily extreme-weather decision-support framework for winter and overlap crops in Australia and the United States. Daily weather observations are transformed into interpretable growing-season indicators, including rainfall totals, dry spells, heavy-rain events, heat exposure, cold/frost exposure, growing degree days, and radiation summaries. These indicators are merged with soil and crop-region-year yield data to predict yield in tonnes per hectare and to support crop-region suitability analysis.

Models were screened with a forward-time validation design: 1989–2010 for training, 2011–2015 for model selection, and 2016–2021 for final testing after refitting on 1989–2015. In the combined Australia and U.S. overlap-crop task, the validation-selected weather model achieved a test RMSE of 0.661 t/ha and a test R^2 of 0.718, while a previous-year yield baseline slightly outperformed it with a test R^2 of 0.720. Beyond prediction, the framework identifies crop-region combinations with the largest observed yield changes, compares weather indicators between high-yield and low-yield years, and ranks candidate crops under the same observed region-year weather and soil profile. The results show that daily extreme-weather features can support interpretable yield prediction and practical crop-choice screening, although temporal persistence is a strong benchmark. Findings are interpreted as predictive associations and planning signals rather than causal effects.

Keywords: crop yield prediction; extreme weather; interpretable machine learning; crop suitability; Australia; United States; decision support.

1 Introduction

Crop-yield prediction is a practical agricultural problem because crop performance depends on the interaction among weather, soil, crop type, location, and management. Machine learning and deep learning methods have been widely used for this task, especially when weather, soil, remote sensing, or multi-source predictors are available [7, 21, 16]. For planning purposes, however, prediction accuracy alone is not enough. Farmers, analysts, and policy users also need to know which crops are more suitable in specific regions, which crop-region combinations are improving or declining, and which weather conditions are associated with favorable or unfavorable yield years.

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This study therefore moves beyond a conventional machine-learning leaderboard. The objective is to build a decision-support framework that predicts yield in tonnes per hectare and translates model outputs into interpretable agricultural signals. This emphasis follows recent work warning that black-box yield models should be interpreted carefully when used to discuss climate and agricultural planning questions [15, 14]. The analysis focuses on winter and overlap crops in Australia and the United States, using a harmonized May–October growing-season proxy. Daily weather records are converted into interpretable extreme-weather indicators, including seasonal rainfall, dry spells, heat exposure, cold/frost indicators, heavy-rain events, growing degree days, and radiation summaries. These indicators are combined with soil variables and crop-region-year yield records.

The analysis is organized around a sequence of practical questions. First, it tests whether daily extreme-weather and soil features can predict crop yield under a forward-time evaluation, which is closer to the way a decision-support model would be used for future seasons. It then moves from overall prediction to crop-specific evidence by identifying which crop-region combinations show the strongest yield increases or decreases. To make the predictions explainable, the study compares high-yield and low-yield years within selected crop-region pairs and identifies the weather indicators that most clearly separate them. Finally, it uses the trained model to compare candidate crops under the same observed region-year weather and soil profile, producing a weather-conditioned crop-suitability signal in tonnes per hectare.

The main contribution is an interpretable workflow that connects daily weather extremes, soil context, yield prediction, and crop-region suitability. The results are reported as predictive associations and planning signals, not causal effects.

2 Related Work

Machine learning for crop-yield prediction has been reviewed extensively, with recent surveys showing that weather, soil, remote sensing, and historical yield variables are common inputs for both classical machine learning and deep learning models [7, 19, 21]. Large-scale and regional studies further show that yield prediction can be framed at county, region, or country scale when field-level management data are not available [8, 9, 10, 18].

Comparative crop-yield studies commonly evaluate multiple regression and ensemble models because no single algorithm is consistently best across crops, regions, feature sets, and evaluation splits [11, 12, 13, 20]. More recent work also emphasizes multi-source feature design, including weather, soil, satellite, and other environmental inputs [17, 16, 22, 26]. For future extensions, UAV and remote-sensing reviews provide a natural path beyond the current tabular weather-soil frame [23].

Interpretability is especially important when yield models are used to discuss climate or weather impacts. Explainable crop-yield studies warn that black-box predictions can be misleading without careful validation and domain-aware explanation [15, 25]. Transfer-learning and domain-shift studies also show that models trained in one data domain may degrade when applied to another, motivating the cautious cross-country framing used in this study [24, 14].

3 Data and Study Scope

The unit of analysis is a crop-region-year observation. The target variable is yield in tonnes per hectare. The current manuscript focuses on overlap winter crops available in both countries: Barley, Canola, Oats, and Wheat. Australia is represented by New South Wales, Queensland, South Australia, Tasmania, Victoria, and Western Australia. The United States is represented by Colorado, Illinois, Iowa, Kansas, Minnesota, Montana, Nebraska, North Dakota, Oklahoma, South Dakota, Texas, and Washington. The yield and weather data are based on public agricul-

tural and climate data sources, including ABARES crop statistics, USDA NASS Quick Stats, SILO climate data, and NASA POWER daily weather products [1, 3, 4, 2, 5, 6].

All annual yield records are harmonized to tonnes per hectare. Weather variables are kept in physical units: rainfall in mm, temperature in degrees Celsius, and radiation in MJ/m². Soil variables are used as region-level persistent context and include available water capacity, bulk density, clay, sand, silt, pH, cation-exchange capacity, organic-carbon-related fields where available, and depth indicators across standard soil-depth layers. Rows with missing yield are removed before modeling. Numeric predictors are median-imputed inside the modeling pipeline, while categorical predictors are imputed with the most frequent training value.

The weather-feature table contains 594 rows because it has one row for each country-region-year combination: 18 regions multiplied by 33 years from 1989 to 2021. The main overlap-crop model frame contains 1,946 rows because each country-region-year weather row is joined to multiple crop-specific yield records. The merge key for daily-weather features is country-region-year; the modeling target is crop-region-year yield, so the same region-year weather profile can be paired with several crops grown in that region-year.

Table 1: Dataset scope used in the daily extreme-weather decision-support study.

dataset	rows	countries	regions	crops	years	unit	crop_scope
Extreme-weather feature table	594	2	18		1989-2021	country-region-year daily-weather features	
Australia overlap crop model frame	789	1	6	4	1989-2021	crop-region-year yield with weather and soil	Barley, Canola, Oats, Wheat
U.S. overlap crop model frame	1157	1	12	4	1989-2021	crop-region-year yield with weather and soil	Barley, Canola, Oats, Wheat
Australia + U.S. overlap crop model frame	1946	2	18	4	1989-2021	cross-country overlap crops	Barley, Canola, Oats, Wheat
Australia + U.S. wheat model frame	591	2	18	1	1989-2021	wheat-only cross-country comparison	Wheat

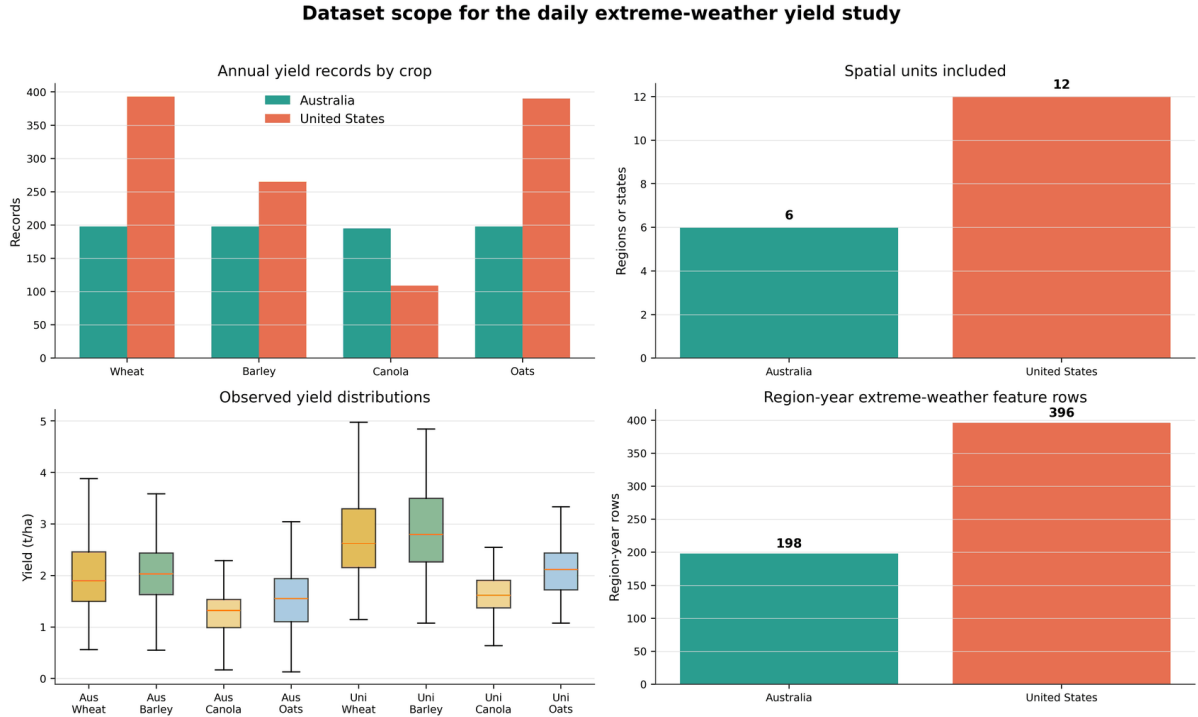


Figure 1: Dataset scope and modeling target. The figure summarizes the country, crop, region, yield, and region-year weather-feature coverage used to construct the crop-region-year modeling frame.

4 Feature Engineering

Daily weather records are aggregated into growing-season indicators using May–October as the common seasonal window. The feature set includes rainfall totals and means, dry-day counts,

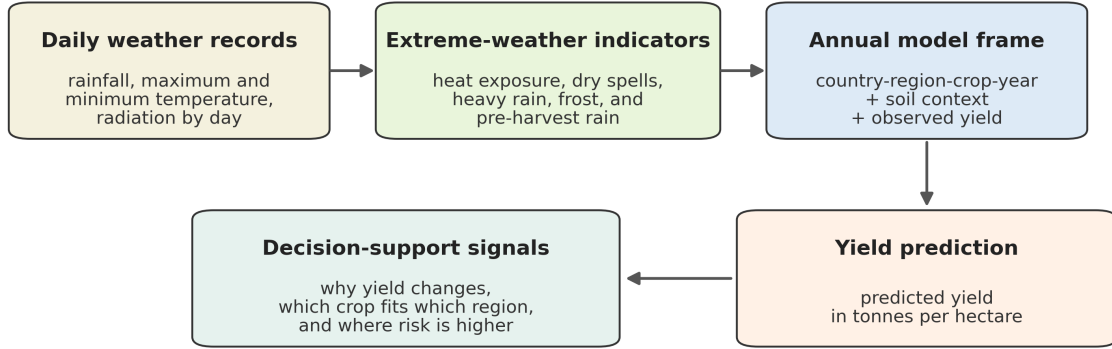
maximum dry-spell length, heavy-rain intensity indicators, heat-day counts, cold and frost exposure, growing degree days, and seasonal radiation summaries. This design is consistent with prior yield-prediction studies showing that weather and environmental inputs are central predictors in both classical machine-learning and deep-learning yield models [7, 16, 21]. These variables are intentionally interpretable so that model outputs can be explained in agronomic language.

Soil variables are added as persistent regional context. Weather features represent year-specific seasonal stress, while soil features represent slower-changing growing conditions. The main modeling comparison evaluates both weather-only and weather-plus-soil feature sets, following the broader multi-source feature-design direction used in recent crop-yield modeling work [17, 22, 26].

Table 2: Grouped definitions of the daily extreme-weather indicators used to construct the crop-region-year model frame.

Indicator group	Computation / threshold	Output
Seasonal rainfall	Sum and mean of daily rainfall over May–Oct.	rain_sum ; rain_mean
Radiation	Sum and mean of daily solar radiation over May–Oct.	radiation_sum ; radiation_mean
Growing degree days	Sum of $\max(Tmean_d - 5^\circ\text{C}, 0)$. If missing, $Tmean_d = (Tmax_d + Tmin_d)/2$.	degree-days
Seasonal windows	Repeat selected rainfall, dry, heat, frost, heavy-rain, and radiation features for early, mid, and late season.	early = May–Jun; mid = Jul–Aug; late = Sep–Oct
Dry exposure	Dry days use $\text{rain} < 1$ mm and $\text{rain} < 2$ mm. Dry spells are the longest consecutive dry run. Long dry-spell events count runs ≥ 7 or ≥ 14 days using $\text{rain} < 1$ mm.	days; event count
Heavy-rain exposure	Count days with $\text{rain} \geq 10, 20, 25$, or 50 mm/day. Also compute maximum rolling 1-, 3-, and 7-day rainfall.	days; mm
Heat exposure	Heat days use $Tmax_d \geq 30, 35$, or 40°C . Heatwave events count runs of at least 3 days at 30 or 35°C .	days; event count
Heat degree days	Sum of $\max(Tmax_d - 30^\circ\text{C}, 0)$ and $\max(Tmax_d - 35^\circ\text{C}, 0)$.	$^\circ\text{C}$ -days
Hot-dry days	Count days where $Tmax_d \geq 30^\circ\text{C}$ and $\text{rain} < 1$ mm.	days
Frost and cold exposure	Frost days use $Tmin_d \leq 0^\circ\text{C}$; cold days use $Tmin_d \leq 5^\circ\text{C}$. Frost events require at least 2 consecutive frost days.	days; event count
Pre-harvest rain risk	Use the last 7, 14, and 21 days before Oct 31. Storm flag = 1 if last-21-day max 3-day rain ≥ 25 mm or last-14-day heavy-rain count ≥ 1 .	mm; count; binary

From daily weather to practical crop-yield signals



Daily rows are summarized before modelling, so the model unit remains interpretable: country-region-crop-year.

Figure 2: Daily extreme-weather feature design. Daily weather observations are aggregated into interpretable growing-season indicators, merged with soil context, and used for yield prediction, crop-region suitability, and weather-conditioned crop comparison.

5 Modeling Strategy

The model set includes linear and nonlinear learners: Ridge regression, ElasticNet, Random Forest, Extra Trees, Histogram Gradient Boosting, and a multilayer perceptron neural network. Comparing multiple model families is useful because crop-yield studies often find that performance depends on the crop, region, feature set, and evaluation design rather than on one universally best algorithm [11, 12, 13, 20]. Model selection uses a forward-time validation protocol. Candidate models are first trained on 1989–2010 and compared on 2011–2015. The selected model and feature set are then refitted on 1989–2015 and evaluated once on the final 2016–2021 test period. This design better represents future-season prediction than a random split and avoids choosing the best model directly from the test set [14, 24].

Model performance is evaluated with MAE, RMSE, and R^2 . MAE and RMSE are reported in tonnes per hectare. Two simple baselines are also evaluated for the combined overlap-crop task: a historical crop-region mean baseline and a previous-year yield baseline. Feature-group ablation compares crop-region-year only, rainfall-only, temperature-only, radiation-only, extreme-event-only, all-weather, and weather-plus-soil feature sets. Interpretation is based on feature-group ablation, high-versus-low yield comparisons, and weather-conditioned crop rankings. This emphasis on post-model interpretation follows recent explainable-AI work warning that black-box yield models require careful, domain-aware explanation before being used for climate or planning conclusions [15, 25].

For weather-conditioned crop suitability, candidate crops are interpreted as model-based screening alternatives rather than direct planting recommendations. Crop comparisons are strongest when both the best crop and the runner-up have historical records in the same region. The reported crop-choice table is therefore filtered to region-observed candidate crops. These rankings do not account for cultivar choice, planting calendar, irrigation, fertilizer, disease pressure, input cost, or market constraints.

Table 3: Validation-selected model for each modeling task. Models are selected on 2011–2015 validation performance, refitted on 1989–2015, and reported on the 2016–2021 test period.

Dataset	Feature set	Selected model	Validation R^2	Test R^2	Test RMSE	Test rows
Australia + U.S. overlap crops	Extreme-weather only	ExtraTrees	0.777	0.718	0.661	353
Australia + U.S. wheat	Extreme-weather only	ElasticNet	0.667	0.689	0.697	105
Australia overlap crops	Extreme-weather + soil	HistGradientBoosting	0.602	0.490	0.890	144
U.S. overlap crops	Extreme-weather only	ExtraTrees	0.851	0.801	0.534	209

Table 4: Baseline and feature-group ablation for the combined Australia and U.S. overlap-crop task. Models are selected by validation performance, then refitted and evaluated on the 2016–2021 test period.

Feature / baseline	Selected model	Test R^2	Test RMSE
Historical crop-region mean	Mean baseline	0.604	0.784
Previous-year yield	Lag-1 baseline	0.720	0.659
Crop + region + year only	ExtraTrees	0.703	0.679
Rainfall only	ExtraTrees	0.746	0.628
Temperature only	ExtraTrees	0.751	0.622
Radiation only	ExtraTrees	0.711	0.670
Extreme events only	ExtraTrees	0.733	0.644
Weather all	ExtraTrees	0.718	0.661
Weather + soil	HistGradientBoosting	0.729	0.648

6 Results

6.1 Predictive Performance

The validation-based time-split evaluation shows that daily extreme-weather indicators contain useful information for crop-yield prediction, but it also shows that simple crop-region temporal baselines are strong. In the combined Australia and U.S. overlap-crop task, the validation-selected weather model was ExtraTrees with the extreme-weather-only feature set, reaching a test RMSE of 0.661 t/ha and a test R^2 of 0.718. This performance was competitive with, but slightly below, the previous-year yield baseline on the final test set. The U.S. overlap-crop model achieved the strongest held-out performance, with ExtraTrees reaching a test RMSE of 0.534 t/ha and a test R^2 of 0.801. The Australia-only overlap-crop task was more difficult, with a validation-selected Histogram Gradient Boosting model obtaining a test R^2 of 0.490. For the cross-country wheat-only task, the validation-selected model was ElasticNet, with a test RMSE of 0.697 t/ha and a test R^2 of 0.689.

The ablation results clarify where the predictive signal comes from. For the combined overlap-crop task, the historical crop-region mean baseline achieved a test R^2 of 0.604, and the previous-year yield baseline achieved a test R^2 of 0.720. Among weather groups, temperature-only and rainfall-only models reached test R^2 values of 0.751 and 0.746, respectively. The weather-plus-soil setting reached a test R^2 of 0.729 but was not selected as the best validation model. These results suggest that daily weather features are informative, especially temperature and rainfall summaries, but that crop-region history and temporal persistence remain important benchmarks. The main value of the weather-feature workflow is therefore not simply beating a persistence baseline, but providing interpretable planning signals about weather-conditioned yield risk and crop suitability.

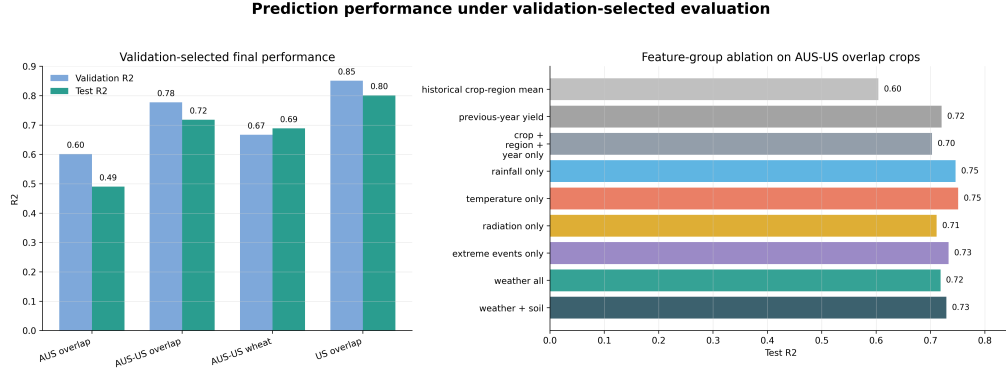


Figure 3: Time-split model performance and feature-group ablation. Bars report validation and test R^2 values for the validation-selected model in each task, and the ablation panel reports test R^2 for baseline and feature-group settings in the combined Australia and U.S. overlap-crop task.

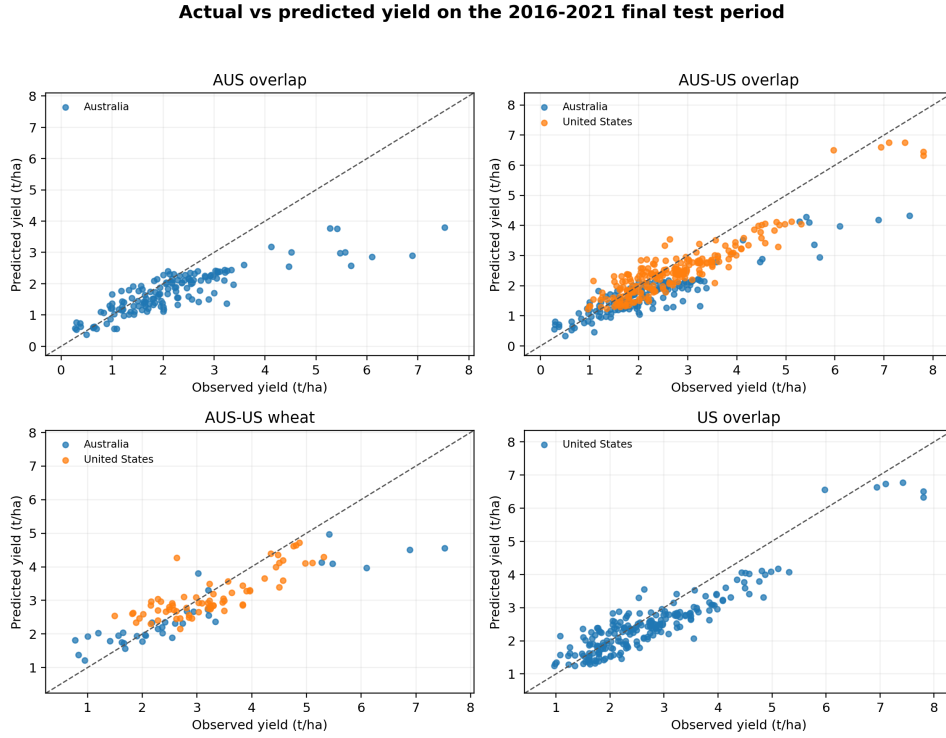


Figure 4: Observed versus predicted yield in the held-out test period. Each point represents a crop-region-year observation in 2016–2021, allowing visual assessment of model calibration in the original yield scale.

6.2 Crop-Region Yield Change

Observed yield changes were crop-specific and region-specific. The largest observed gain occurred for Australia Tasmania Wheat, where mean yield increased by 1.986 t/ha from 1989–2000 to 2010–2021. The next largest gain was United States Colorado Barley, with an increase of 1.923 t/ha. The largest observed decreases were Australia New South Wales Oats, which declined by 0.350 t/ha, and United States Washington Oats, which declined by 0.270 t/ha.

Table 5: Selected crop-region yield gains and losses between early and recent periods.

country	region	crop	early yield, 1989-2000	recent yield, 2010-2021	change, t/ha	status
Australia	Tasmania	Wheat	3.535	5.522	1.986	increase
United States	Colorado	Barley	5.107	7.030	1.923	increase
Australia	Tasmania	Barley	2.559	4.066	1.507	increase
United States	Minnesota	Wheat	2.587	3.767	1.179	increase
United States	Illinois	Wheat	3.396	4.545	1.149	increase
Australia	Tasmania	Canola	1.125	2.174	1.049	increase
United States	Iowa	Wheat	2.707	3.609	0.902	increase
United States	North Dakota	Wheat	2.075	2.882	0.807	increase
Australia	New South Wales	Oats	1.464	1.114	-0.350	decrease
United States	Washington	Oats	2.526	2.256	-0.270	decrease
United States	South Dakota	Barley	2.385	2.222	-0.163	stable
Australia	New South Wales	Canola	1.525	1.446	-0.079	stable
Australia	Queensland	Oats	0.684	0.639	-0.045	stable
United States	Montana	Oats	1.835	1.805	-0.030	stable

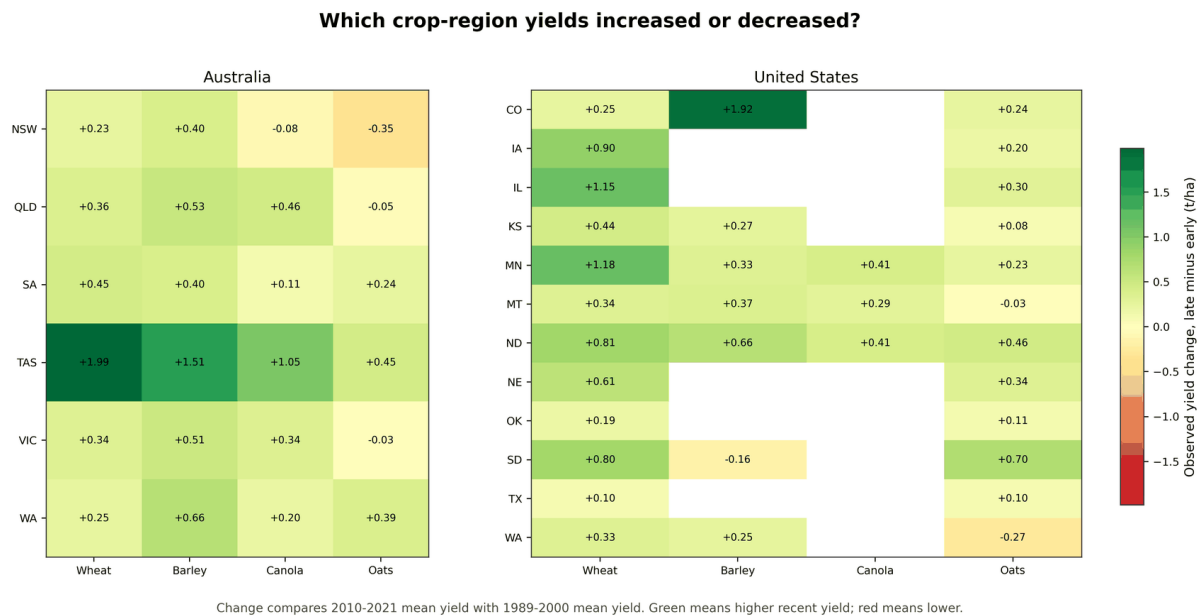


Figure 5: Observed yield change by crop and region. The heatmap compares mean yield in 1989–2000 with mean yield in 2010–2021, identifying crop-region combinations with the largest increases or decreases.

6.3 Weather Signatures of High-Yield Years

The high-versus-low yield comparison translates model results into weather language. For each crop-region pair, the analysis compares the highest predicted-yield years with the lowest predicted-yield years and records which weather indicators differ most. This comparison is used as an interpretability layer rather than as causal attribution, consistent with cautions in explainable yield modeling [15, 14]. The strongest high-low gap was found for United States Colorado Barley, where the predicted difference between high-yield and low-yield years was

1.667 t/ha. High-yield years in that crop-region pair were associated with higher mean minimum temperature, higher growing degree days, higher mean maximum temperature, and a longer dry spell.

A second strong example was Australia New South Wales Wheat, with a predicted high-low gap of 1.624 t/ha. High-yield years were associated with higher seasonal rainfall, higher daily rainfall, higher maximum three-day rainfall, and lower seasonal radiation. These statements should be interpreted as predictive associations within the dataset, not causal effects.

Table 6: Weather signatures separating high-predicted-yield years from low-predicted-yield years. The gap is the predicted yield difference within each crop-region pair.

Country	Region	Crop	Gap	Weather indicators in high-yield years
United States	Colorado	Barley	1.667	Higher mean minimum temperature (+0.6 C), higher growing degree days (+114.3 degree-days), higher mean maximum temperature (+0.7 C), and longer dry spell (+4.1 days).
Australia	New South Wales	Wheat	1.624	Higher seasonal rainfall (+255.1 mm), higher daily rainfall (+1.4 mm/day), higher maximum 3-day rainfall (+27.8 mm), and lower seasonal radiation (-185.1 MJ/m ²).
Australia	New South Wales	Barley	1.492	Higher daily rainfall (+1.1 mm/day), higher seasonal rainfall (+194.7 mm), lower late-season radiation (-113.4 MJ/m ²), and lower seasonal radiation (-167.7 MJ/m ²).
Australia	Tasmania	Wheat	1.420	Lower late-season radiation (-56.8 MJ/m ²), lower seasonal radiation (-80.7 MJ/m ²), higher growing degree days (+70.3 degree-days), and higher mean maximum temperature (+0.4 C).
Australia	Victoria	Wheat	1.389	Higher seasonal rainfall (+163.6 mm), higher daily rainfall (+0.9 mm/day), more late-season rainfall (+29.0 mm), and higher maximum 3-day rainfall (+24.7 mm).
Australia	Victoria	Barley	1.334	Lower late-season radiation (-97.9 MJ/m ²), fewer hot-dry days (-3.7 days), higher seasonal rainfall (+122.7 mm), and higher daily rainfall (+0.7 mm/day).

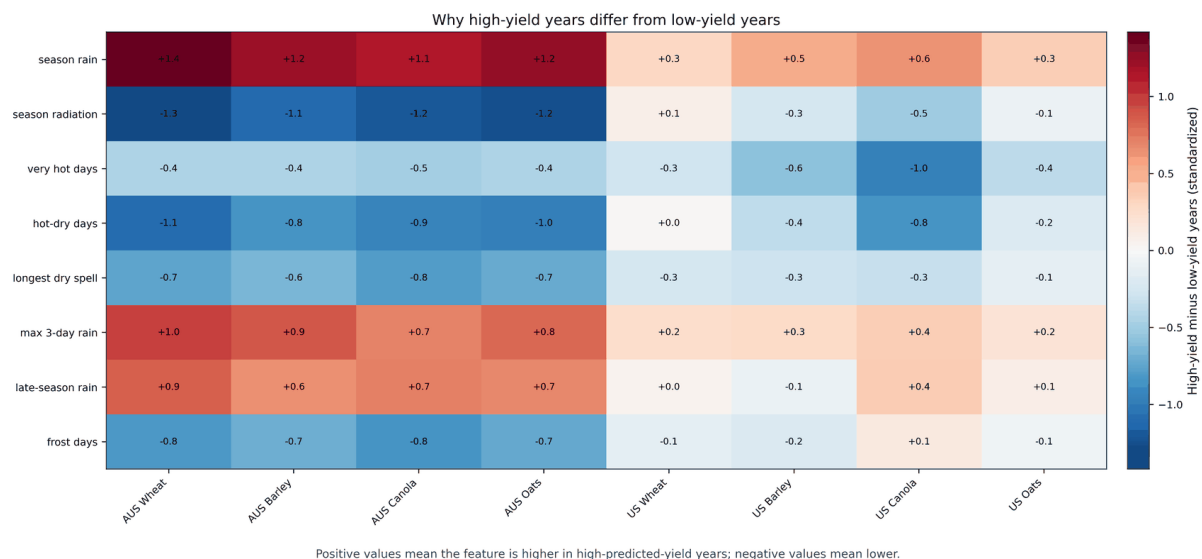


Figure 6: Weather signatures of high-yield years. For selected crop-region pairs, the figure compares weather indicators in high-predicted-yield years with low-predicted-yield years. Values should be interpreted as predictive associations rather than causal effects.

6.4 Weather-Conditioned Crop Suitability

The crop-choice analysis asks which crop is predicted to perform best under the same observed region-year weather and soil profile. This turns yield prediction into a more practical planning signal and aligns with recent work framing yield modeling as a decision-support problem rather than only an accuracy benchmark [8, 16, 21]. To reduce unsupported extrapolation, the reported crop-choice table uses candidate crops with historical records in the same region. The largest region-filtered crop-choice advantage occurred in United States Colorado in 2010, a dry-spell stress year. Under that observed profile, Barley was predicted to outperform Wheat by 3.778 t/ha, with a best predicted yield of 6.707 t/ha. Another strong case was United States Iowa in 2006, where Wheat was predicted to outperform Oats by 1.643 t/ha under a moderate-season profile.

Table 7: Largest weather-conditioned crop-choice advantages under observed region-year weather and soil profiles. Candidate crops are restricted to crops with historical records in the same region, and at most one case is shown per region.

country	region	year	weather regime	best crop	runner-up crop	advantage, t/ha	best yield, t/ha	candidates
United States	Colorado	2010	Dry-spell stress	Barley	Wheat	3.778	6.707	3
United States	Iowa	2006	Moderate season	Wheat	Oats	1.643	4.378	2
United States	Illinois	2013	Wet/storm risk	Wheat	Oats	1.600	4.256	2
Australia	Tasmania	2008	Cold/frost stress	Wheat	Barley	1.196	4.030	4
United States	Washington	2003	Dry-spell stress	Wheat	Barley	0.958	3.583	4
United States	North Dakota	2018	Cold/frost stress	Barley	Wheat	0.900	3.725	4
United States	Montana	1991	Cold/frost stress	Barley	Wheat	0.791	2.776	4
Australia	New South Wales	1991	Wet/storm risk	Wheat	Oats	0.764	2.216	4
United States	Texas	2011	Hot-dry stress	Barley	Wheat	0.701	2.182	3
Australia	Victoria	2016	Wet/storm risk	Wheat	Barley	0.624	3.892	4

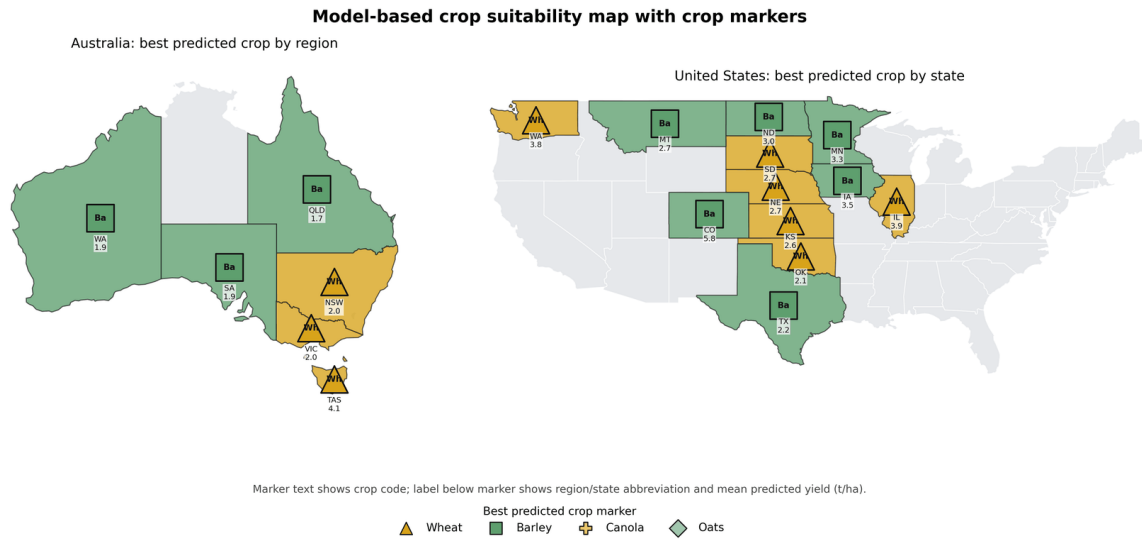


Figure 7: Model-based crop suitability map. For each region, the map shows the crop with the highest predicted yield under the regional weather and soil profile. Markers are decision-support signals, not prescriptive recommendations.

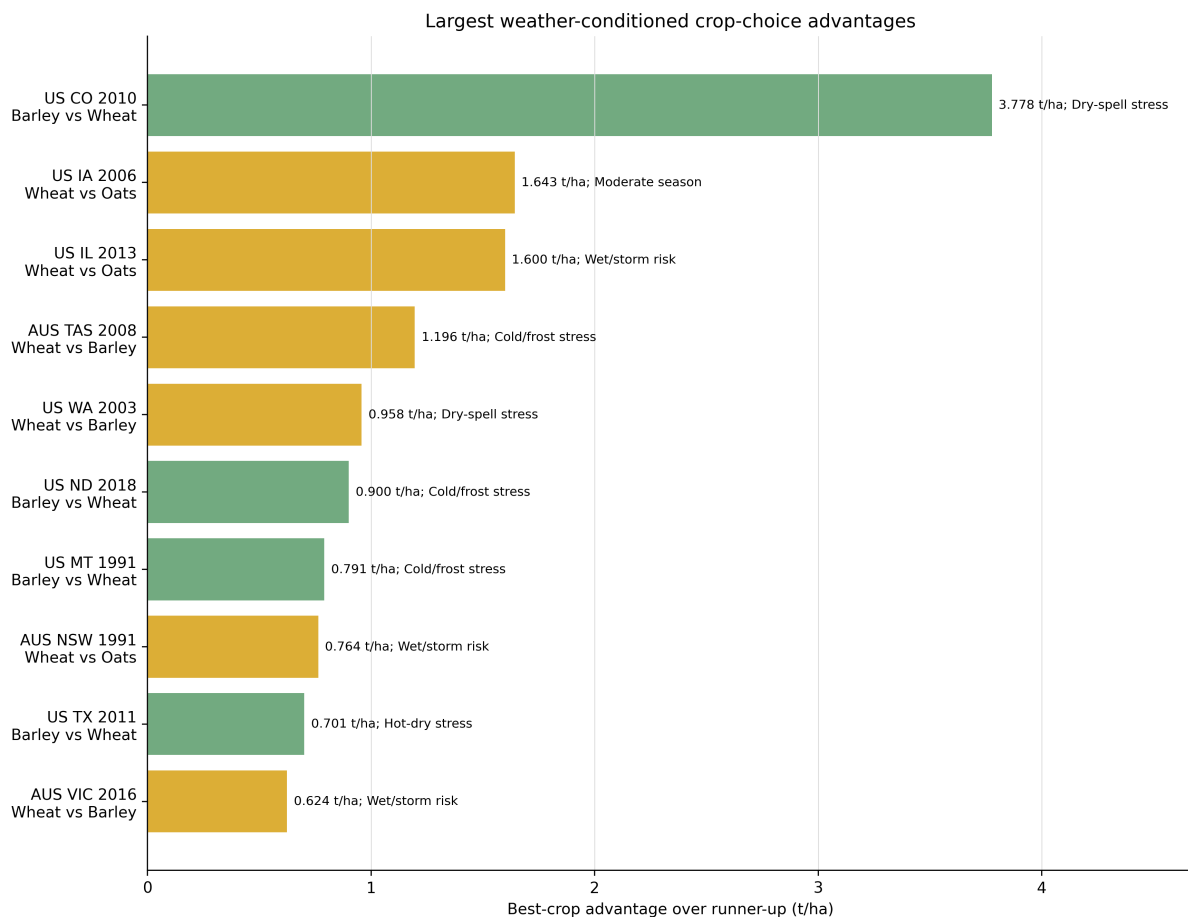


Figure 8: Weather-conditioned crop-choice advantage. The figure highlights region-year cases where the best predicted crop has the largest yield advantage over the runner-up crop under the same weather and soil conditions.

7 Discussion

The results show that daily extreme-weather features are not only predictive, but also useful for explaining crop-region performance in operational terms. This supports the broader direction of multi-source and interpretable crop-yield modeling, where weather, soil, and environmental context are used to move from prediction toward actionable agricultural information [19, 22, 25]. The framework identifies where yield change is concentrated, which weather indicators distinguish favorable and unfavorable years, and which crops are predicted to be more suitable under the same region-year weather and soil profile.

The crop-choice analysis should be interpreted as a screening layer rather than a direct prescription. It can help identify candidate crops or crop-region combinations for closer agronomic evaluation, but it does not replace local knowledge about irrigation, market demand, cultivar choice, disease risk, input costs, or management constraints. This caution is especially important because yield models can degrade when moved across time, geography, or data domains [14, 24].

8 Limitations and Future Work

This analysis is predictive and associative, not causal. The May–October growing season is a harmonized proxy and does not capture crop-specific calendars. Region-level data cannot fully

represent within-region farm management, irrigation, cultivar, fertilizer, disease, or local soil variation. Weather products also differ across countries, with Australia and the United States relying on different source products. Future work should add crop-specific calendars, finer spatial resolution, management variables where available, and uncertainty intervals for crop-choice rankings. Remote sensing and UAV-based data streams are also promising extensions for improving spatial detail beyond the current tabular weather-soil frame [23, 24].

9 Conclusion

This study presents an interpretable daily extreme-weather framework for crop-yield prediction and crop-region suitability analysis. The validation-selected combined Australia and U.S. overlap-crop weather model achieved competitive held-out predictive performance, although the previous-year yield baseline slightly outperformed it on the final test set. The post-model analysis translated predictions into crop-specific and region-specific planning signals by identifying yield gains and losses, weather signatures of high-yield years, and weather-conditioned crop-choice advantages. These outputs make the approach more useful for agricultural planning than a conventional yield-prediction leaderboard alone.

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